

**BUDDY: An Intelligent Robot Dog for Offline Voice
Interaction, Visual Perception, and Safety-Aware
Multimodal Control**

INDIVIDUAL THESIS

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Emotion-Aware Interaction, Human Fall Detection, and Adaptive Object
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DECLARATION AND SUPERVISOR CERTIFICATION

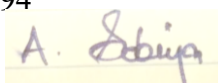
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ABSTRACT

This individual thesis presents the visual-intelligence contribution to BUDDY, an intelligent robot-dog designed for multimodal human-robot interaction. The original visual scope focused on face recognition, facial-expression interpretation, and camera-based face following. Following supervisory feedback that face recognition alone provided insufficient research depth, the component was expanded into a broader adaptive visual-intelligence subsystem incorporating unknown-person handling, household enrollment, emotion-aware responses, adaptive object learning, human-fall detection, live camera monitoring, and a Flutter-based companion application. The implemented system combines OpenCV-based face detection, hybrid recognition mechanisms, DeepFace-based seven-category expression analysis, MobileNetV2 feature extraction for user-taught objects, and MediaPipe pose landmarks for fall-state reasoning. A Flask API links the vision pipeline to the mobile application for live video, notifications, enrollment, media capture, and status presentation. Functional validation used approximately five known users, around twenty unfamiliar image/video samples for unknown-person testing, and three learned everyday objects: a mug, box, and phone. Face recognition, live emotion analysis, unknown-person detection and enrollment, object learning, camera streaming/recording, and pan-tilt face tracking were demonstrated at subsystem level. Occasional false rejection of enrolled users as unknown was observed, particularly under variations in view and image quality. Fall-detection logic is implemented, while final full-body physical validation remains pending because the available development-camera view was unsuitable. The work demonstrates how a companion robot can move beyond isolated face recognition toward an extensible, privacy-conscious visual-awareness layer that supports interaction, safety, and personalized behaviour.

Keywords: companion robot, face recognition, facial expression recognition, adaptive object learning, human fall detection

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
FER	Facial Expression Recognition
HTTP	Hypertext Transfer Protocol
LBPH	Local Binary Patterns Histograms
ML	Machine Learning
ROS 2	Robot Operating System 2
TTS	Text-to-Speech
UI	User Interface

1 INTRODUCTION

1.1 Overall project context

Companion robots are embodied interactive systems expected to perceive people and their surroundings, communicate naturally, and perform physical actions in a shared human environment. BUDDY was developed as a low-cost intelligent robot-dog prototype that combines three complementary research streams: offline auditory interaction, visual and facial intelligence, and environment-aware navigation and physical control. The complete platform uses a Raspberry Pi 4 and ROS 2 to coordinate sensing, perception, interaction, and movement while maintaining a safety boundary between high-level requests and physical motor authority [1]-[3].

Within this multimodal architecture, visual intelligence is essential because the robot must do more than receive commands. A companion robot should be able to notice whether a person is present, distinguish familiar and unfamiliar users, orient its camera toward a user, interpret observable facial-expression cues, and use visual context to support safe and personalised interaction. Visual perception therefore operates both as a human-robot interaction channel and as an environmental awareness channel.

The overall BUDDY research emphasizes modularity. Voice processing, visual processing, navigation, safety supervision, and actuation are separated so that each can be developed and tested independently before final integration. This structure is particularly important for the visual subsystem because computationally intensive image processing must coexist with speech processing, sensor monitoring, application communication, and physical actuation on resource-constrained hardware.

1.2 Individual component context

The individual component addressed in this thesis is the visual-intelligence subsystem led by Sobiya Anton Suresh. The original scope, recorded in the project paper, focused on visual and facial intelligence: face tracking, face recognition, and seven-class facial-expression classification. During project review, supervisory

feedback indicated that face recognition alone did not provide sufficient individual research depth. The component was therefore broadened after further investigation to include adaptive object learning, unknown-person awareness and enrollment, human-fall detection, emotion logging, camera media services, and a companion application that exposes visual information to the user.

This evolution changed the research question from whether BUDDY could recognise a face to whether an affordable companion robot could maintain a useful visual-awareness layer that supports recognition, affect-aware interaction, safety-related observation, incremental learning, and user-facing monitoring. The subsystem consequently combines conventional computer-vision techniques, pre-trained deep-learning models, local feature databases, event logic, and mobile/web application integration.

The visual work also includes the two-axis camera head. A detected face position is compared with the camera-frame centre and translated into pan/tilt corrections. The project paper physically demonstrated camera-based face following, while the current development branch extends the software around that capability with richer recognition and application features.

1.3 Problem statement

A companion robot that only reacts to explicit voice commands has limited awareness of the person and environment in front of it. Basic face detection can indicate that a face exists, but it does not identify whether the person is known, provide an interpretable expression cue, distinguish a trusted household member from an unfamiliar visitor, remember a user-taught object, or identify a possible fall. For BUDDY to behave as a meaningful companion rather than a remotely controlled mobile platform, the visual subsystem must provide persistent and actionable context.

At the same time, real-world visual recognition is unreliable under changes in illumination, camera angle, blur, partial occlusion, and distance. The project experienced this directly: an enrolled user could occasionally be classified as UNKNOWN even though recognition was generally functional. A practical design

must therefore handle uncertainty rather than assume every single-frame prediction is correct. It must also operate with limited computing resources and integrate with a live camera stream and application without causing unacceptable processing lag.

The individual research problem is therefore to design and implement an extensible visual-intelligence subsystem for an embedded companion robot that combines familiar-person recognition, facial-expression interpretation, unknown-person handling, adaptive object learning, fall-detection logic, active face tracking, and user-facing application services while remaining responsive and transparent about uncertainty.

1.4 Research gap

Prior research provides strong methods for isolated face verification, face embeddings, facial-expression recognition, lightweight image feature extraction, and body-pose estimation [4]-[9]. However, these techniques are often evaluated as standalone machine-learning tasks. A companion robot requires them to operate together under practical constraints: a continuously changing video stream, limited compute, user enrollment, event persistence, hardware actuation, and an interface through which a household user can see and respond to visual events.

The gap addressed by this work is therefore primarily an integration and adaptation gap. Instead of proposing a new foundational deep neural network, the research investigates how established visual methods can be assembled into a coherent, extensible perception layer for a low-cost robot-dog. Particular attention is given to uncertainty handling, incremental user/object enrollment, asynchronous processing, and the boundary between implemented capability and validated physical behaviour.

1.5 Aim and objectives

The aim of this individual research is to develop and evaluate an adaptive visual-intelligence subsystem for BUDDY that supports person recognition, facial-expression-aware interaction, active face tracking, unknown-person awareness, adaptive object learning, and safety-oriented fall detection while integrating with the companion application and the wider robot architecture.

The objectives are: (1) implement familiar-person recognition with local enrollment and unknown-person rejection; (2) analyse live facial-expression cues across seven categories and expose the result to the robot/application; (3) support pan/tilt camera-head following of a detected face; (4) implement an unknown-person event workflow that can capture, notify, review, and enroll a new household member; (5) enable a user to teach everyday objects to BUDDY and later identify them through learned visual features; (6) implement pose-based fall-detection logic suitable for later full-body validation; (7) provide camera streaming, snapshot, recording, media retrieval, and visual status through a Flutter application; and (8) evaluate the subsystem functionally while documenting observed limitations rather than reporting unsupported accuracy statistics.

1.6 Research questions

RQ1: How can a low-cost companion robot combine face recognition, facial-expression analysis, active tracking, unknown-person handling, adaptive object learning, and fall-related visual reasoning within a single visual subsystem?

RQ2: How can recognition uncertainty be managed so that occasional low-confidence or ambiguous frames do not immediately become trusted identity decisions or repeated security events?

RQ3: Can the visual subsystem provide useful, observable outputs to a user-facing application while remaining responsive enough for live camera operation and robot interaction?

RQ4: What practical limitations emerge when these functions are tested using development cameras and a small, project-scale group of users rather than a large controlled benchmark?

1.7 Scope limitations and assumptions

The research is scoped to functional prototype validation rather than population-level biometric or clinical evaluation. Approximately five known people were used during familiar-person trials, while around twenty unfamiliar image/video samples were used to exercise unknown-person handling. These observations are sufficient to

demonstrate functional behaviour but are not sufficient for a generalised recognition-accuracy claim.

Facial-expression output is treated as an observable expression category, not as a definitive measurement of a person's internal emotional state. The current fall-detection logic is implemented using pose landmarks, but final full-body physical validation is pending because the available development-camera view did not provide a suitable full-body field of view. Object-learning validation used three everyday objects - a mug, a box, and a phone - and should therefore be understood as proof of functional adaptive learning rather than broad object-recognition benchmarking.

The current application operates successfully with Sobiya's subsystem backend. Full deployment with every final BUDDY subsystem on the completed robot is treated separately as integration work. Some dashboard telemetry values in the prototype backend are demonstrative rather than direct physical sensor measurements and are not used as research results in this thesis.

1.8 Individual contribution

Sobiya Anton Suresh was responsible for the visual-intelligence research stream and for the primary implementation of the Flutter companion application. Her work includes face and expression processing, local face enrollment and memory, unknown-person event handling, object learning, pose-based fall-detection logic, emotion logging/reporting, visual API endpoints, live camera/media services, and the application screens that consume these services. S. S. Y. Wickramasinghe contributed to UI design direction for the application, while Ashandth Uthayashankar assisted with final integration and troubleshooting of the physical camera-head mechanism. These integration contributions do not change ownership of the visual-intelligence component.

GitHub Copilot was used as a programming-assistance tool during parts of application development. Generative AI assistance used in thesis preparation is disclosed separately in Appendix B. Technical claims in this thesis are based on

project source code, project documents, supplied screenshots, and stated functional tests.

1.9 Thesis structure

Chapter 2 reviews relevant work in social/companion robotics, face recognition, facial-expression recognition, active visual tracking, lightweight transfer learning, and pose-based fall detection. Chapter 3 explains the research and validation methodology. Chapter 4 presents requirements and the subsystem design. Chapter 5 describes implementation, integration, and testing. Chapter 6 reports the observed results. Chapter 7 discusses contributions and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and identifies future work. References, a glossary, and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was organised around technologies required by the individual component rather than around a single model family. The main themes were companion-robot interaction, face recognition and embeddings, facial-expression recognition, lightweight visual feature extraction for embedded systems, body-pose estimation, and system integration for robot perception. Priority was given to peer-reviewed papers already used by the BUDDY project and to foundational works directly related to the implemented methods.

The review distinguishes between algorithm-level performance reported in the literature and performance demonstrated by the BUDDY prototype. Published benchmark results provide technical context but are not transferred to BUDDY as if they were measured locally. This distinction is important because the project uses different cameras, users, lighting conditions, enrollment procedures, and computational constraints.

2.2 Domain and theoretical foundations

Companion and socially assistive robots are expected to combine embodiment, sensing, autonomy, and interaction rather than functioning as isolated software agents. Ahmed et al. identify human-robot companionship as an emerging area in which anthropomorphic and zoomorphic forms can support engagement, while Clabaugh and Mataric emphasise the importance of autonomy in socially assistive robotics [1], [2]. For BUDDY, a dog-inspired embodiment motivates visual behaviours such as looking toward a person, recognizing familiar users, and adapting interaction to visible cues.

Modern face-recognition systems commonly transform an input face into a compact representation and compare representations using a similarity measure. DeepFace and FaceNet established influential deep-learning approaches for face verification and embedding-based recognition [4], [5]. These works demonstrate the value of learned representations, but practical robot deployment still requires face detection,

image-quality checks, identity thresholds, enrollment storage, and rejection logic for unknown users.

Facial-expression recognition is typically formulated as image classification over categories such as happy, sad, angry, fear, surprise, disgust, and neutral. Reviews of the field show strong progress with convolutional neural networks, but also persistent sensitivity to illumination, pose, occlusion, identity variation, and uncontrolled environments [6]. Importantly, a predicted facial-expression label is not equivalent to a direct measurement of internal emotional state. In this thesis the output is therefore treated as an interaction cue rather than a psychological diagnosis.

For embedded visual learning, MobileNetV2 provides a computationally efficient convolutional architecture based on inverted residuals and linear bottlenecks [8]. Its relatively lightweight feature representation makes it suitable as a fixed feature extractor for a project in which a user teaches a small number of objects without retraining a large network from scratch. For body-pose estimation, BlazePose provides 33 body keypoints designed for real-time operation on mobile-class hardware [9], supporting geometric reasoning over shoulder, hip, and head landmarks for fall-state heuristics.

2.3 Related research

Taigman et al. demonstrated that deep neural representations could substantially improve face verification, while Schroff et al. proposed FaceNet, which maps face images directly into an embedding space where distance corresponds to identity similarity [4], [5]. These methods motivate the use of embedding-style comparison in BUDDY. However, the current prototype also retains simpler local mechanisms because deployment reliability and ease of enrollment are more important than reproducing a large-scale benchmark architecture.

Ekundayo and Viriri's review of facial-expression recognition identifies deep learning as a dominant approach while highlighting dataset, pose, and environment challenges [6]. The BUDDY research paper initially described a seven-class CNN operating on 48 x 48 grayscale faces. The current integrated development branch instead uses DeepFace's facial-attribute analysis interface where available, with a

neutral fallback if analysis fails. This evolution illustrates a pragmatic engineering decision: the research goal is a working multimodal subsystem, not merely preservation of one earlier classifier implementation.

Sandler et al.'s MobileNetV2 work demonstrates a design targeted at efficiency across mobile and embedded tasks [8]. BUDDY uses an ImageNet-pretrained MobileNetV2 backbone without the classification head to extract feature vectors from a central object region. Multiple observations are averaged to produce a stored representation, and later frames are matched using similarity. This is closer to interactive few-shot memorisation than to general-purpose object detection.

Bazarevsky et al. present BlazePose as a real-time pose-tracking method capable of predicting a 33-keypoint body topology [9]. BUDDY's fall-detection extension uses MediaPipe Pose and then applies interpretable geometric rules over landmarks. The approach is deliberately heuristic: it checks landmark visibility, body scale, and body orientation and requires a possible-fall condition to persist before a fall state is confirmed. This reduces dependence on a separately trained fall-classification dataset but requires a camera view that contains enough of the body.

2.4 Existing systems, products and techniques

Commercial companion and home-monitoring products commonly separate interaction, security monitoring, and camera functionality into different devices or subscriptions. BUDDY instead explores whether these functions can coexist in a single low-cost experimental robot. From a research perspective, the closest technical building blocks are not one specific competing robot but a combination of face-recognition libraries, facial-attribute frameworks, mobile feature extractors, pose trackers, and robot middleware.

DeepFace provides a convenient abstraction for facial-attribute analysis and supports common face-recognition workflows [7]. OpenCV provides the image acquisition, face detection, preprocessing, image encoding, recording, and display operations used throughout the prototype. MediaPipe supplies pose landmarks for the fall module. Flutter is used for the user-facing application, while Flask exposes HTTP endpoints from the Python visual backend. ROS 2 remains the wider robot

integration framework, allowing the visual component to coexist with voice, safety, sensing, and motor subsystems [3].

A key difference between a desktop demonstration and a companion robot is continuous operation. A robot must decide how often to run expensive inference, how to prevent repeated alerts, how to recover when a model is unavailable, and how to expose results to other components. The current implementation therefore uses frame skipping, cached detections, asynchronous threads, fallback behaviours, persistent JSON stores, and alert cooldowns. These engineering techniques are part of the practical research contribution because they enable the algorithms to function together.

2.5 Gap synthesis and conceptual framework

The literature establishes strong individual methods but does not remove the integration problem faced by a low-cost companion robot. The conceptual framework adopted in this thesis is therefore a layered visual-awareness pipeline: acquire a frame; assess whether useful visual targets are present; recognise or reject identity; infer an expression cue; update tracking and application state; perform optional adaptive learning or safety reasoning; and only then expose an event or robot response.

Three principles guide the design. First, uncertain identity should fail toward UNKNOWN rather than force a possibly incorrect known-person label. Second, repeated observations should be used where possible to stabilise decisions, as seen in consecutive unknown counts, alert cooldowns, multiple enrollment frames, and multiple object-learning frames. Third, user-facing actions such as enrollment should remain explicit: an unfamiliar face is presented for review rather than silently becoming a trusted household identity.

3 RESEARCH METHODOLOGY

3.1 Research design

This research follows a design-and-build methodology with iterative functional evaluation. The individual contribution was developed as a software-intensive subsystem within a larger embodied robot. Requirements were refined from the project goal, supervisory feedback, implementation constraints, and observed behaviour during testing. New functions were added incrementally and tested first at subsystem level before being connected to the mobile application and physical camera-head mechanism.

The evaluation approach is intentionally functional rather than statistical. The available participants and test samples were small and opportunistic, so the research does not report a population-level biometric accuracy, facial-expression accuracy, sensitivity/specificity for fall detection, or benchmark object-recognition score. Instead, each capability is evaluated by whether the intended end-to-end behaviour can be observed under representative project conditions and by documenting failure modes encountered during use.

3.2 Data, participants and study materials

Known-person testing involved approximately five real individuals: Sobiya's hostel friends and Ashandth Uthayashankar. Unknown-person behaviour was additionally exercised using around twenty images and video samples of unfamiliar people displayed from a phone. These samples were used to verify rejection and event handling rather than to construct a formal biometric benchmark.

Adaptive object learning was tested with three common objects: a mug, a box, and a phone. For a learning session, the software collects multiple frames of the target object and stores an averaged visual feature representation. The box example is documented by a supplied screenshot showing a stored learned-object match with confidence displayed in the live interface.

The main study materials consisted of a laptop/webcam development setup, the BUDDY camera/head subsystem, the project Python codebase, the Flutter

companion application, JSON stores for known faces, emotion logs and learned objects, and the project robot hardware. The final fall-detection module requires a camera position that captures sufficient full-body landmarks; the available development view was not suitable for a complete physical fall trial at the time of thesis preparation.

Material / evidence	Use in study	Status
~5 known users	Familiar-person recognition and enrollment	Functionally exercised
~20 unfamiliar image/video samples	Unknown-person rejection and alert behaviour	Functionally exercised
Mug, box, phone	Adaptive object learning/recognition	Functionally exercised
Laptop/webcam and BUDDY camera	Live visual testing and development	Used
Flutter application + Flask API	User-facing monitoring, enrollment, media	Working at subsystem level
MediaPipe fall module	Pose-based fall reasoning	Implemented; final full-body physical validation pending

Table 3.1 Validation materials and evidence sources

3.3 Experimental and study procedure

Face functionality was tested by enrolling known users and observing live recognition labels under ordinary development conditions. Unfamiliar samples were then presented to verify that the system could display UNKNOWN rather than incorrectly assigning a stored identity. The app-level unknown-person workflow was tested by allowing the backend to create an unknown event, presenting the captured face in the notifications interface, and exposing an action through which a name could be entered and the person added to the household dataset/database.

Emotion functionality was exercised during live face processing. The dominant expression and confidence were displayed alongside the identity label, stored for recognised users at a controlled logging interval, and made available to the

application. The implementation also includes reaction mappings for different expression categories and a robot-reaction interface; final physical behaviour depends on the wider hardware integration.

For object learning, the user initiates a learning action and names the object. The module gathers approximately thirty observations from a central crop, extracts MobileNetV2 features, normalises and averages them, and stores the representation in the local object database. Later frames are compared with stored representations. Successful examples were observed for a mug, box, and phone.

For fall detection, MediaPipe Pose identifies body landmarks. The algorithm checks landmark visibility and apparent body size, estimates body orientation using shoulder and hip midpoints, marks a potential fall when the body becomes sufficiently horizontal, and requires persistence for approximately two seconds before confirmation. The module was implemented and code-tested, but final full-body physical validation remains a planned post-submission integration task because of the development camera's restricted field of view.

3.4 Measures and evaluation metrics

The primary measures are functional outcomes rather than aggregate statistical metrics. For person recognition, the observed outcome is known-user recognition versus UNKNOWN rejection. For unknown-person handling, the outcomes are event persistence, captured face availability, application notification, and successful enrollment path. For emotion analysis, the outcome is availability of a dominant category and confidence value during live operation. For adaptive object learning, the outcome is successful storage and later matching of a user-taught object. For camera services, the outcomes are successful live streaming, snapshot/recording operation, and media retrieval.

The system uses internal thresholds as engineering controls. The face-recognition configuration includes similarity/confidence thresholds; the unknown-person workflow requires multiple consecutive unknown observations and applies an alert cooldown; the object-learning module uses a similarity threshold of 0.70; and the fall detector requires a possible-fall condition to persist for about two seconds. These

thresholds are implementation parameters and should not be confused with measured accuracy values.

3.5 Data analysis

Results were analysed descriptively. Screenshots, console outputs, application behaviour, stored JSON artefacts, and successful/unsuccessful functional observations were used to determine whether each feature completed its intended flow. Particular attention was given to negative evidence and boundary conditions. For example, the occasional classification of an enrolled user as UNKNOWN is retained as a limitation rather than removed from the report.

No inferential statistical analysis was performed because the test sample is too small and was not collected using a controlled sampling protocol. Confidence values displayed by models are reported only as model outputs in individual examples; they are not interpreted as calibrated probabilities of correctness.

3.6 Ethics, privacy, security and responsible AI

The subsystem processes face images, identity labels, expression logs, and unknown-person captures. These are sensitive in the context of a household companion robot. The prototype therefore favours local storage and explicit enrollment. Unknown people are not automatically assigned a trusted identity; the application presents the event for human review before enrollment. In a production system, additional controls would be required for consent, retention periods, encryption, access control, deletion, and compliance with applicable privacy law.

Facial-expression analysis is treated cautiously because visible expression and internal emotion are not identical. The system uses expression categories as interaction cues only. Fall detection is likewise a safety-assistance feature, not a medical diagnosis or emergency-service replacement. These boundaries are important to prevent overclaiming the capability of prototype AI.

Development used GitHub Copilot as coding assistance and generative AI was used during thesis drafting and language refinement. The academic-integrity disclosure in

Appendix B identifies these uses and states that project-specific technical claims were checked against code, project documents, and test evidence.

3.7 Reproducibility and research artefacts

The implementation is maintained in the BUDDY GitHub repositories. The visual branch contains configuration, database management, emotion detection/logging, face recognition, fall detection, frame management, object learning, robot bridging/control, shared state, and utility modules. Persistent development artefacts include `known_faces.json`, `emotion_logs.json`, and `object_database.json`. The separate Flutter repository contains screens for home status, camera, emotion, notifications, household management, control, activity, settings, and related functions.

The repository also contains placeholder model files for a TensorFlow face model. Because those files are zero-byte placeholders in the inspected branch, this thesis does not claim that the model binary itself is reproduced from the repository. The current integrated runtime instead retains hybrid recognition paths, including local database comparison and LBPH memory fallback. This distinction improves reproducibility by separating available source artefacts from unavailable trained-model content.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The visual-intelligence subsystem was designed around both interaction and safety-awareness requirements. It needed to run with a live camera, recognise enrolled users, reject unfamiliar users, expose a seven-category facial-expression cue, track a face with the pan/tilt head, support incremental object learning, implement pose-based fall reasoning, and provide user-visible status without blocking the camera loop.

Non-functional requirements included responsiveness, modularity, local operation where practical, graceful degradation if an analysis library failed, transparent UNKNOWN handling, limited storage growth, and compatibility with the wider Python/ROS 2 project. Because the subsystem is part of an embodied robot, processing decisions should not independently bypass the project's safety architecture for physical movement.

ID	Requirement	Priority
FR1	Detect and recognise enrolled faces from a live camera feed	High
FR2	Reject low-confidence or unfamiliar faces as UNKNOWN	High
FR3	Analyse observable facial expression across seven categories	High
FR4	Track a detected face using pan/tilt head orientation	High
FR5	Generate an unknown-person event and support explicit enrollment	High
FR6	Learn and later identify user-taught everyday objects	Medium
FR7	Implement pose-based fall-state reasoning	Medium

ID	Requirement	Priority
FR8	Stream, capture, record and list camera media in the app	Medium
NFR1	Avoid blocking the live camera loop during logging/network tasks	High
NFR2	Preserve local, reviewable data stores for enrollment and learned objects	Medium

Table 4.1 Functional requirements for the visual-intelligence subsystem

4.2 Component architecture within the complete solution

The subsystem can be viewed as five cooperating layers. The acquisition layer provides camera frames. The perception layer detects faces, recognises identity, analyses expression, extracts object features, and estimates body pose. The state and persistence layer stores known faces, emotion history, learned objects, and recent unknown events. The service layer exposes Flask endpoints and a live MJPEG stream. The presentation layer is the Flutter application. A robot-integration layer connects visual outputs to pan/tilt control, shared robot state, speech, and emotion-related reactions.

This layered design limits coupling. For example, the application does not need direct access to OpenCV objects; it requests JSON status or a video endpoint. Similarly, the face recogniser does not directly implement the mobile enrollment UI. This separation made it possible to test the visual subsystem independently on a laptop before connecting it to the full robot.

4.3 Design alternatives and decisions

The project moved from a narrower recognition-only design toward a hybrid visual-intelligence design because supervisory feedback indicated that simple face recognition was insufficient as an individual research contribution. This led to extensions that add safety, adaptation, and user interaction rather than only improving one recognition classifier.

For identity, the design retains more than one recognition path. The code supports a trained TensorFlow model when available, a local embedding/database comparison, and an LBPH face-memory fallback. Although maintaining multiple paths increases complexity, it improves resilience during development and supports enrollment without requiring complete model retraining.

For emotion analysis, the integrated branch uses DeepFace where available and falls back to neutral if the analysis library fails. This prioritises continuity of robot operation. For object learning, MobileNetV2 feature extraction was chosen instead of training a detector from scratch because the desired interaction is to teach a small number of household objects quickly. For fall detection, pose geometry was selected because it provides interpretable body-orientation cues and avoids the need to collect a dedicated fall-video training dataset for the prototype.

Decision	Alternative considered	Rationale
Hybrid face recognition	Single classifier only	Supports enrollment, fallbacks, and development resilience
DeepFace integrated emotion analysis	Only the earlier custom CNN	Rapid live integration and seven-category output
MobileNetV2 feature memory	Training an object detector from scratch	Efficient transfer features and user-taught objects
MediaPipe Pose + geometric rules	Dedicated fall-action classifier	Interpretable logic and no custom fall dataset required
Repeated unknown observations	Alert on first unknown frame	Reduces transient false alerts
Flask API + Flutter client	Monolithic desktop-only UI	Separates visual backend from user-facing interface

Table 4.2 Principal design decisions and rationale

Figure 4.1 summarises the principal visual-intelligence methods used in the subsystem and the reason each was selected for BUDDY. The term “model” in the development summary is used broadly: OpenCV Haar cascades and local cosine/LBPH matching are classical computer-vision or pattern-matching techniques,

while DeepFace, MobileNetV2, and MediaPipe BlazePose provide pre-trained deep-learning-based capabilities.

Model / Method	What it does	Purpose in BUDDY
OpenCV Haar Cascade	Detects human faces	Fast face detection from live camera
Cosine Similarity / LBPH	Compares stored face patterns	Recognizes known users and identifies unknown users
DeepFace	Analyzes facial expressions	Detects emotions such as happy, sad, angry, neutral, etc.
MobileNetV2	Extracts image features	Learns and recognizes user-taught objects such as mug, box and phone
MediaPipe BlazePose	Tracks body landmarks	Supports posture analysis and fall detection

Table 4.3 Visual-intelligence methods and their role in BUDDY

4.4 Detailed design

Face detection uses OpenCV Haar-cascade processing with grayscale conversion and histogram equalisation. A face crop is padded and then validated for minimum dimensions, acceptable brightness, and blur. Poor-quality crops are rejected before identity processing. The recognition module can preprocess a face for a TensorFlow classifier, create a compact handcrafted embedding for the local database, or use the LBPH memory subsystem. Database matching uses cosine similarity and a configurable threshold.

Unknown-person logic is intentionally conservative. A missing identity, explicit UNKNOWN label, or low-confidence result sets the current face state to UNKNOWN. The system then requires a sequence of unknown observations before generating an event. Before sending an alert it performs another database check on the captured face; if no registered identity meets the threshold, the face image is

JPEG-encoded, converted to Base64, timestamped, and made available to the app. A cooldown prevents repeated alerts for the same continuing situation.

Expression analysis calls DeepFace with the emotion action and maps results to angry, disgust, fear, happy, sad, surprise, and neutral. The logger limits how frequently an identified person's expression is stored, reducing redundant records. It can generate summary graphs and supports Telegram reporting when credentials are configured. Expression categories are also mapped to conceptual robot reactions such as changes in eye state, tail-wag state, speed state, or sound state; the physical availability of each reaction depends on the integrated hardware implementation.

The object-learning module loads ImageNet-pretrained MobileNetV2 without the classification head and uses global average pooling to obtain feature vectors from a 224 x 224 object crop. During learning, approximately thirty frame features are collected, averaged, normalised, and written to the object database under the user-provided name. During identification, the current feature vector is compared with stored vectors using a dot-product similarity; the default recognition threshold is 0.70.

The fall module uses MediaPipe Pose with detection and tracking confidence configured at 0.7. Landmark visibility and body area are checked to reject unsuitable or very small targets. Shoulder and hip midpoints provide an estimate of torso orientation. A near-horizontal body configuration is treated as a possible fall, and the condition must remain for about two seconds before the detector reports a confirmed fall. This temporal requirement is intended to reduce false alarms from short transitions.

4.5 Standards and professional practice

Professional practice for this prototype includes modular source control, explicit uncertainty handling, clear separation between demonstrated and planned functionality, and privacy-aware treatment of visual data. The wider BUDDY project uses ROS 2 as the primary robot middleware, while the visual subsystem exposes narrow interfaces rather than directly assuming authority over safety-critical movement [3].

The implementation also follows defensive programming practices visible in the source: exception handling around optional libraries, timeouts for application requests, bounded log storage, repeated confidence checks, and asynchronous background tasks for network or file I/O. These practices are especially important in a robot that must continue operating even when one perception service degrades.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The visual subsystem is implemented primarily in Python and integrates OpenCV, NumPy, TensorFlow/Keras-compatible components, DeepFace, MediaPipe, MobileNetV2, Flask, Flask-CORS, Pandas, Matplotlib, and HTTP/network libraries. The user-facing companion application is implemented in Flutter/Dart. Persistent prototype data is stored in JSON files and captured camera media is stored as local image/video files.

The BUDDY robot platform uses Raspberry Pi 4 hardware and ROS 2 Jazzy in the wider system. During individual development, many visual functions were first exercised on a laptop webcam because this allowed faster iteration. The camera-head mechanism and face-following behaviour were subsequently tested on the physical robot with team integration support.

Technology	Role in Sobiya subsystem
Python	Core visual backend and integration logic
OpenCV	Camera input, face detection, preprocessing, display, media encoding
DeepFace	Live seven-category facial-expression analysis
MediaPipe Pose	Body landmark extraction for fall-detection logic
MobileNetV2	Pretrained feature extractor for adaptive object learning
Flask + CORS	HTTP API and MJPEG video service
Flutter / Dart	Companion application UI and API client
JSON stores	Known faces, emotion logs, learned objects
ROS 2 / shared robot interfaces	Wider BUDDY integration and coordination

Table 5.1 Development technology stack

5.2 Component implementation

The main visual runtime initialises the database manager, emotion detector, emotion logger, face-recognition module, robot controller, optional LBPH face memory, object learner, camera state, and Flask service. The system processes live frames while maintaining cached recognition state so that computationally expensive operations do not have to run on every frame.

The face pipeline performs detection, crop extraction, quality validation, recognition, and expression analysis. For a recognised user, the system updates the shared status, logs expression at a controlled interval, and can trigger the reaction interface asynchronously. Newly recognised people can also receive a spoken greeting through the project's speech function. For an unknown user, the repeated-detection and recheck logic described in Chapter 4 creates an application-visible security event only after sufficient persistence.

Application services include live MJPEG video, snapshots, start/stop video recording, media listing, media deletion, camera reset, unknown-person alert retrieval/clearing, unknown-person enrollment, object-learning initiation, object identification, robot-status reporting, household-member status, settings, security status, and notification retrieval. Some broader robot-control and diagnostic endpoints currently include demonstrative values and are therefore treated as prototype UI integration rather than validated hardware telemetry.

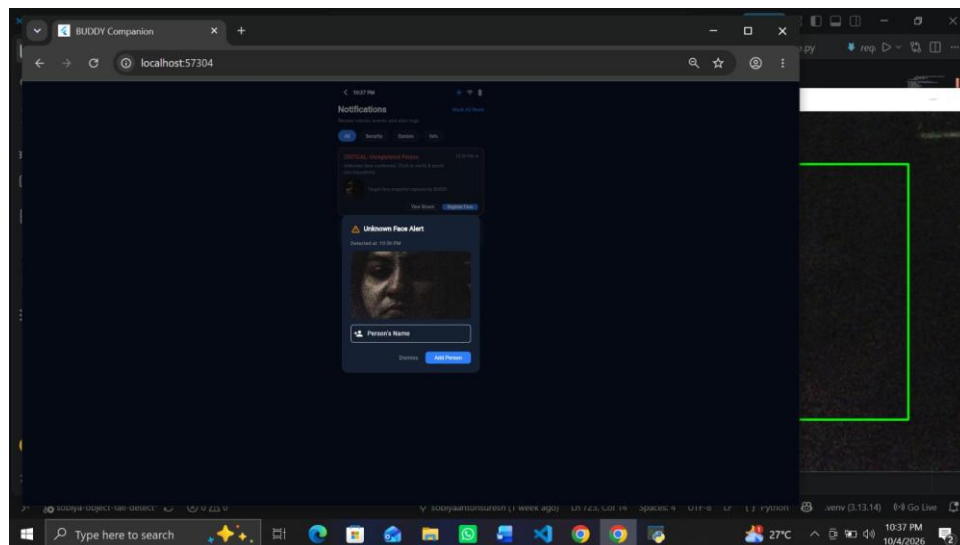


Figure 5.1 BUDDY mobile dashboard showing recognised-user mood and unknown-person status

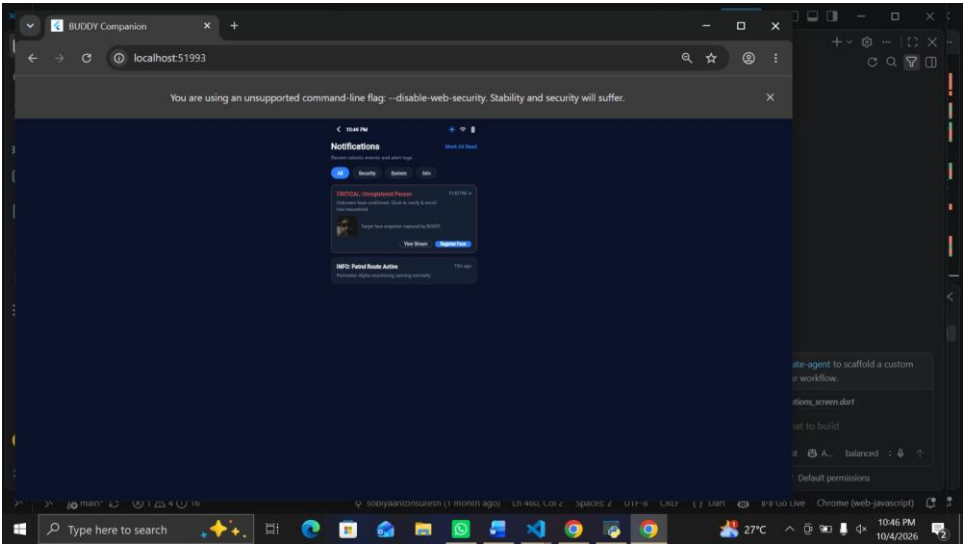


Figure 5.2 Unknown-person security notification and registration action

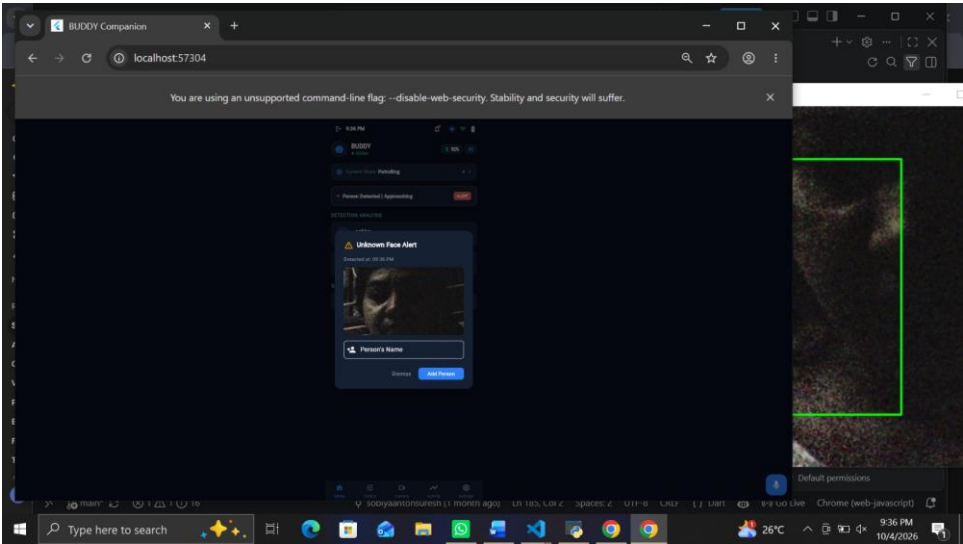


Figure 5.3 Unknown-person enrollment dialog with captured face

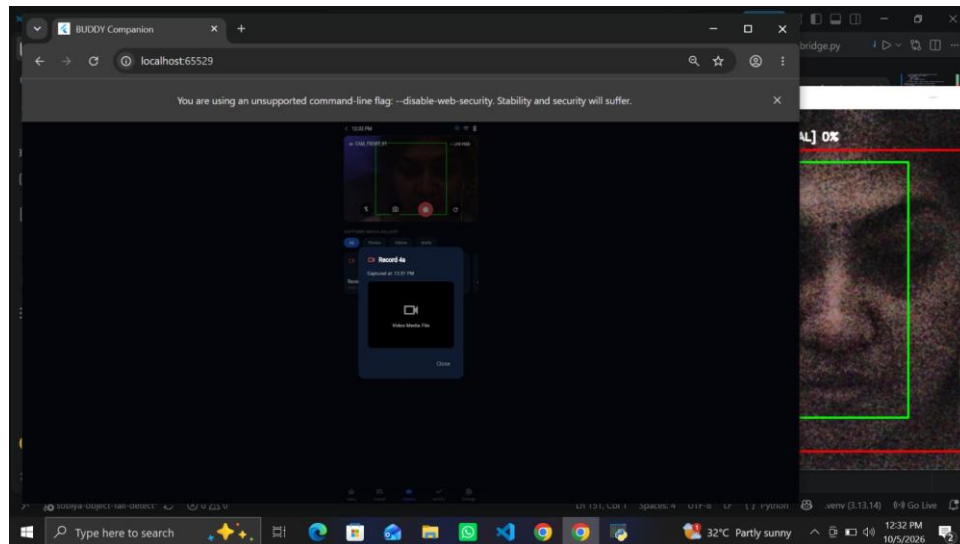


Figure 5.4 Live camera monitoring and recording interface

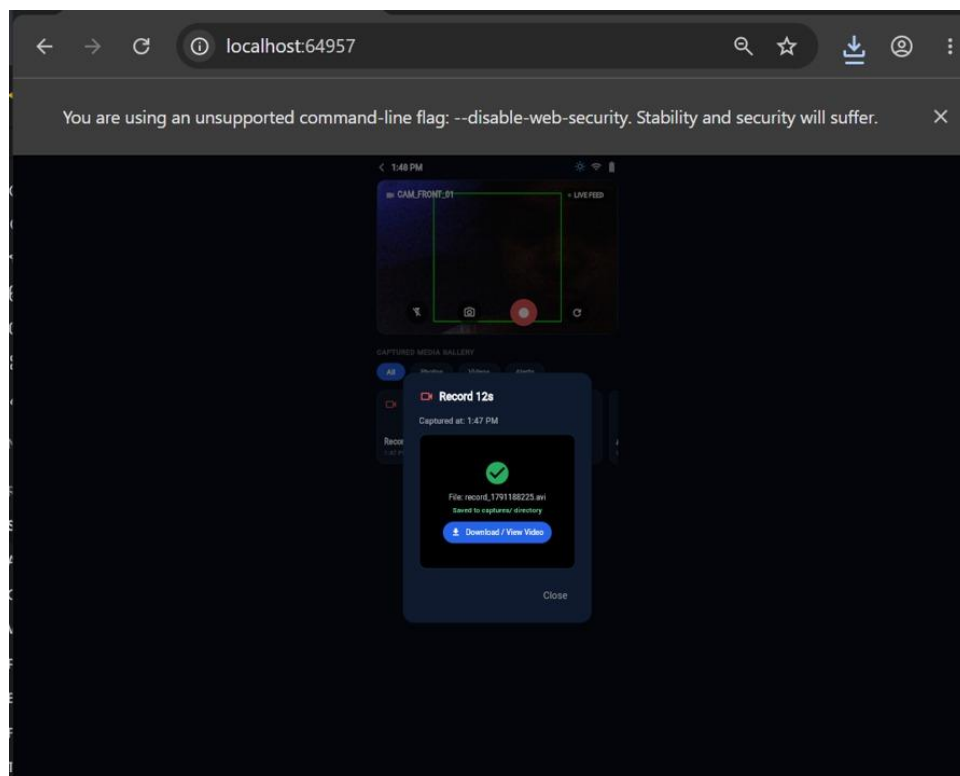


Figure 5.5 Recorded media retrieval interface

5.3 Integration with the other components

At project level, Sobiya's component integrates with Ashandth Uthayashankar's voice/intelligence work and S. S. Y. Wickramasinghe's navigation and physical-control work. The visual subsystem supplies person/expression state and face

position, while the wider robot provides speech, movement, safety supervision, and actuator execution. The project paper physically validated camera-based face following as a perception-to-actuation path.

The physical pan/tilt face-following feature remains part of Sobiya's visual component. Sobiya developed and tested the tracking logic, while Ashandth assisted during final robot integration and troubleshooting. This is recorded as integration support rather than transfer of component ownership. Similarly, Yohan contributed UI design input for the application, while Sobiya implemented the application's functional integration with her backend.

The application is currently working with Sobiya's subsystem independently. Final connection of all application functions to the completed robot stack is an integration task. This separation allows this thesis to claim subsystem functionality without implying that every endpoint has already been validated against final hardware.

5.4 Verification and software testing

Testing combined live demonstrations, repeated manual trials, code inspection, application interaction, and persisted artefacts. Known-face recognition was exercised with approximately five people. Unknown-person handling was tested using around twenty unfamiliar image/video samples as well as live unknown-person scenarios. The results showed that the system can distinguish familiar and unfamiliar cases, although known people can occasionally be rejected as UNKNOWN under unfavourable conditions.

Adaptive object learning was exercised with a mug, box, and phone. The live box example displayed the stored object label and similarity/confidence output after learning. Camera streaming and recording were tested through the application, including successful recording and later media retrieval. Unknown-person events were demonstrated in the application with a captured image and an explicit registration action.

Fall detection is treated differently. The implementation logic is present and the module can run, but the available development-camera positioning did not provide a suitable full-body view for credible physical validation. Consequently, the thesis

records the module as implemented with final full-body validation pending rather than converting an unverified feature into a pass result.

Feature	Evidence / trial	Observed status
Known-person recognition	~5 known users	Working; occasional false rejection to UNKNOWN observed
Unknown-person recognition	~20 unfamiliar image/video samples plus live cases	Working at subsystem level
Unknown alert + enrollment	App notification and captured-face registration flow	Working at subsystem level
Facial-expression analysis	Live seven-category outputs	Working
Pan/tilt face following	Laptop development + physical robot-head test	Working with integration support
Adaptive object learning	Mug, box, phone	Working; learned-object example documented
Camera live stream / recording	Flutter app and Flask media endpoints	Working
Fall detection	Pose module implemented	Final full-body physical validation pending

Table 5.2 Component-level functional test matrix

5.5 Deployment and operation

During development, the Flask backend was commonly accessed locally from the Flutter web build, while mobile builds use a configured local-network IP address. This arrangement confirms application-to-backend behaviour but requires network configuration to be updated for the final robot environment. A production deployment should avoid hard-coded addresses and should use discoverable or configurable service endpoints.

The camera configuration can switch between a development webcam and the robot camera path. For resource efficiency, face detection, emotion processing, and embedding work are performed at configured intervals rather than on every frame. Background threads are used for emotion logging, network operations, and other

work that would otherwise stall the video loop. This architecture was necessary to maintain a usable live visual feed while multiple perception services were active.

6 RESULTS AND VALIDATION

6.1 Research results

The principal research result is that the component progressed from a narrow face-recognition task into a working multi-function visual-awareness layer. The subsystem can maintain identity state, classify unfamiliar faces as UNKNOWN, attach an observable expression category to detected faces, track a face using an actuated camera head, learn a small set of user-named objects, and expose visual events to a companion application.

This broader scope directly addresses the supervisory concern that face recognition by itself was insufficient. The extension demonstrates an iterative research process in which the problem definition evolved after feedback and further investigation. The resulting contribution is not a new universal vision model; it is an integrated, extensible approach tailored to an affordable companion robot.

6.2 Product and technical results

Known-person recognition was demonstrated across approximately five enrolled users. Unknown-person rejection was exercised with roughly twenty unfamiliar images/videos and additional live cases. The system successfully produced UNKNOWN outputs and, in demonstrated application cases, created a critical unregistered-person notification containing the captured face and an action to register the person.

The app-level enrollment flow demonstrates a complete human-in-the-loop trust transition: an unfamiliar face is detected, captured, presented to the user, named through the UI, and added to the local dataset/database. This is preferable to automatically trusting every newly observed face. Live expression output was also demonstrated alongside identity, showing that identity and expression processing can occur in the same frame-processing flow.

Object learning was successfully exercised for a mug, box, and phone. The supplied box example shows the learned label 'THE BOX' with live similarity/confidence output in the low-to-high 80/90 percent range across example frames. These values

are example model outputs, not a measured overall accuracy. The important result is that a user-defined object name was stored and later retrieved by visual similarity.

Camera streaming, recording, media saving, and retrieval were demonstrated through the companion interface. These functions make the visual subsystem observable and usable rather than limiting it to console output.

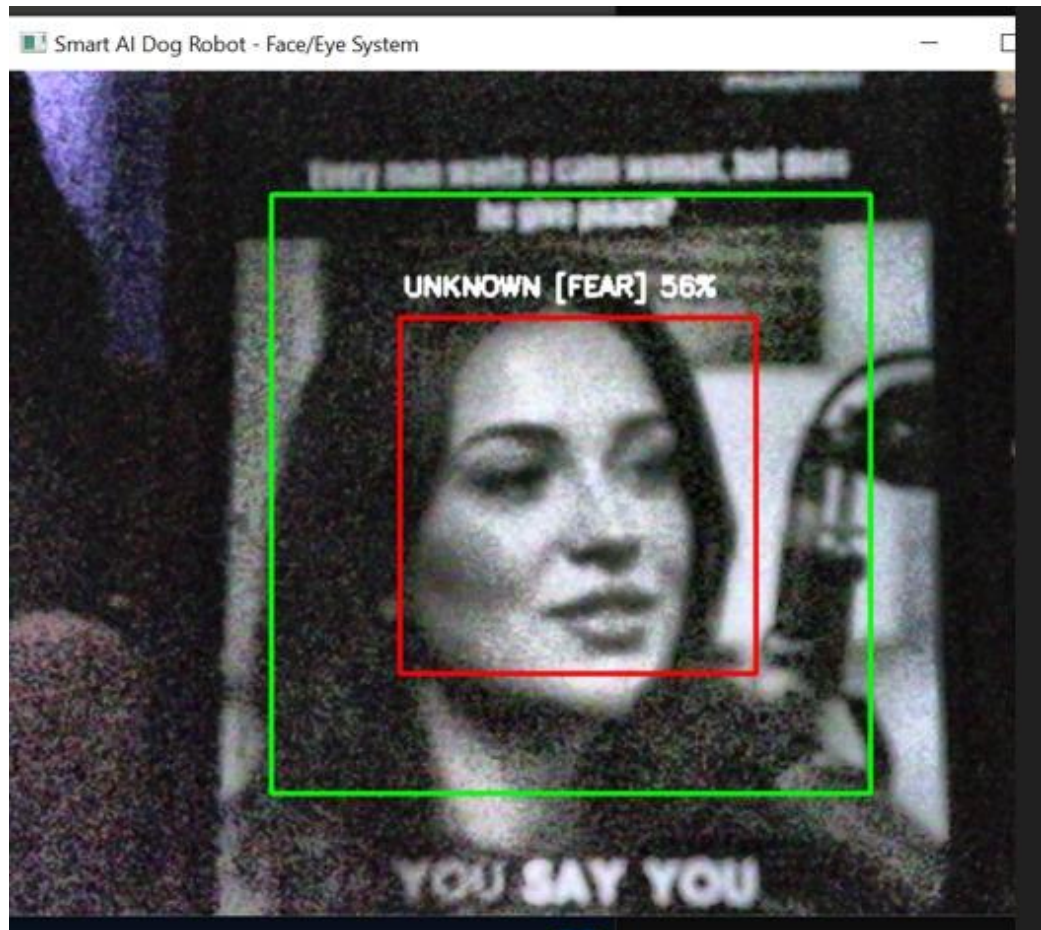


Figure 6.1 Live unknown-person and facial-expression output

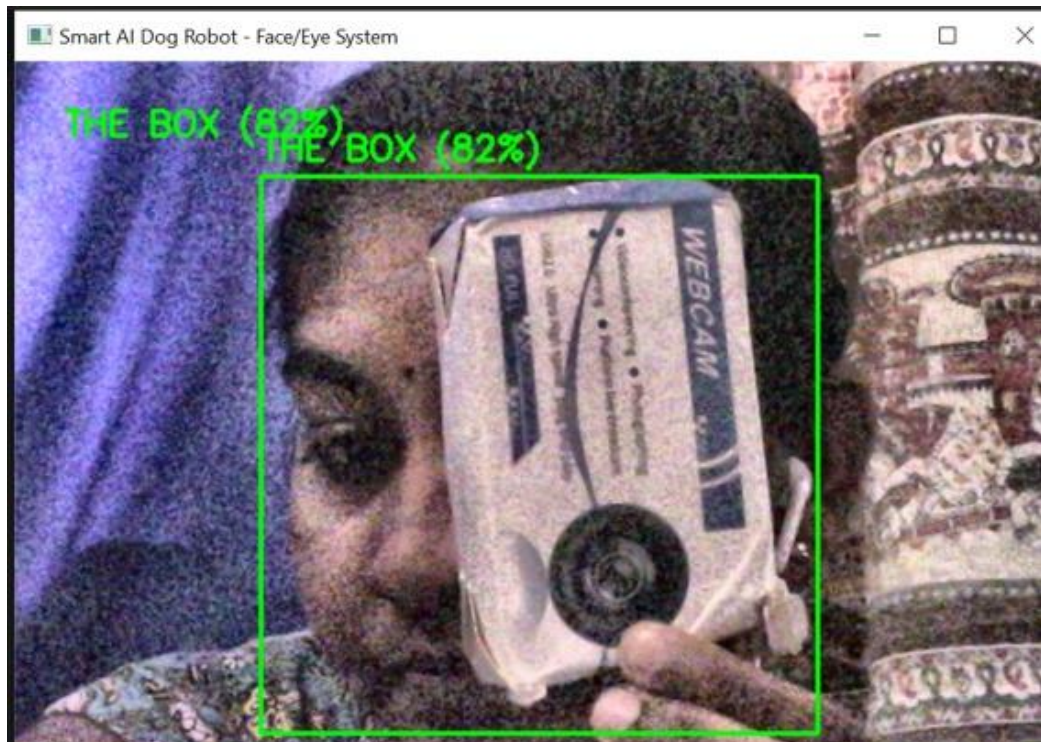


Figure 6.2 Learned-object recognition example for the box

Result area	Observed result	Interpretation
Identity	Known and unknown labels produced in live use	Functional recognition/rejection demonstrated
False rejection	Known user occasionally shown as UNKNOWN	Recognition robustness remains a limitation
Expression	Seven-class expression displayed live output	Functional expression cue available
Security event	Unknown face captured and shown in notifications	End-to-end subsystem alert flow demonstrated
Enrollment	Unknown face can be named and added to household database	Human-reviewed incremental identity learning demonstrated
Object learning	Mug, box, phone learned/tested	Adaptive visual memory demonstrated
Camera media	Live stream, recording, saved media retrieval	User-facing visual monitoring demonstrated
Fall logic	Pose-based detector implemented	Physical full-body validation still required

Table 6.1 Summary of observed subsystem results

6.3 User or stakeholder validation

The principal stakeholder feedback influencing the component came from supervision: the original face-recognition scope was considered too limited, prompting the expansion into additional visual-intelligence features. This feedback materially changed the individual contribution and led to the addition of adaptive object learning, fall-detection logic, security-oriented unknown-person handling, and richer application integration.

Informal user-level validation occurred through the small group of known participants and application use during development. Because no structured usability questionnaire or controlled participant study was completed, this thesis does not report usability scores. The evidence instead demonstrates that the application can present the intended information and actions during real subsystem operation.

6.4 Integration results

Physical face following provides the clearest validated bridge from Sobiya's visual perception to robot actuation. The project paper reports that the camera head physically followed a detected face using pan/tilt actuation. The current visual branch additionally links recognised-user and expression state to a robot-reaction interface and shared state, enabling other components to consume the visual interpretation.

Application integration is working with Sobiya's backend. Full final-robot integration remains an ongoing team activity. This distinction is important: an independently working subsystem is a valid individual result, while complete system integration is a group-level result that should be confirmed only after all components are connected.

6.5 Summary of findings

The evidence supports five main findings. First, multi-function visual awareness is feasible on the BUDDY development stack. Second, conservative unknown-person rejection and user-reviewed enrollment are practical for a household companion prototype. Third, pre-trained feature extraction enables useful interactive object

learning without retraining a full detector. Fourth, real-time integration requires deliberate performance controls such as frame intervals, caching, asynchronous processing, and bounded logging. Fifth, visual robustness remains sensitive to camera conditions, as shown by occasional false rejection of known users and the inability to complete credible full-body fall validation with the available development camera view.

7 DISCUSSION

7.1 Interpretation of findings

The subsystem results show that the value of visual intelligence in a companion robot comes from composition rather than from one classifier in isolation. Recognition provides identity context, expression analysis provides an interaction cue, unknown-person handling supports security awareness, object learning adds adaptation, and the application gives the human user visibility and control over these processes.

The occasional known-to-UNKNOWN error is an important result rather than merely a defect. The system is designed to reject uncertain matches, so a false rejection is safer than confidently assigning the wrong household identity. However, repeated false rejection can reduce usability. Future work should therefore improve enrollment diversity, threshold calibration, image quality handling, and temporal aggregation so that a known user is not judged from a single weak observation.

7.2 Comparison with literature baselines and existing solutions

DeepFace, FaceNet, and related literature demonstrate much stronger benchmark-scale face verification than this small prototype attempts to measure [4], [5], [7]. BUDDY does not reproduce those benchmark conditions and therefore does not claim comparable accuracy. Instead, it adopts the underlying concepts of face representations, similarity, and facial-attribute analysis in an embedded robotic context.

Similarly, MobileNetV2 was designed as an efficient architecture for mobile/embedded visual tasks [8]. BUDDY uses its representation capability in an unconventional interactive-learning workflow: a household user can name an object, allow the robot to gather several observations, and later retrieve that name by similarity. This does not replace a general object detector, but it aligns well with companion-robot personalization.

MediaPipe/BlazePose provides a lightweight foundation for body-pose tracking [9]. The current fall logic uses pose geometry rather than a learned action classifier. This improves interpretability but makes viewpoint and full-body visibility critical. The

inability to validate the feature properly with the existing camera view demonstrates the difference between algorithm availability and system-level evidence.

7.3 Research contribution

The principal individual research contribution is the design of an expanded visual-awareness architecture for a low-cost companion robot. The contribution combines identity, expression, active tracking, adaptive learning, safety-oriented pose analysis, persistent visual memory, event logic, and application integration. It also introduces explicit uncertainty-handling mechanisms rather than treating every frame-level output as reliable.

A second contribution is the documented evolution of the component scope. Supervisory feedback was converted into a research-driven extension rather than simply adding unrelated features. Each extension addresses a distinct limitation of face recognition alone: emotion adds interaction context, unknown-person handling adds trust/security context, object learning adds adaptation, fall detection adds safety context, and the app makes those outputs accessible to the user.

7.4 Product contribution

From a product perspective, Sobiya's work provides the user's visual window into BUDDY. The companion application can show robot state, recognised-person information, expression status, unknown-person events, household enrollment, camera monitoring, and recorded media. This turns the visual subsystem into an interactive feature rather than a hidden research prototype.

The unknown-person enrollment workflow is particularly relevant to a household product concept. Instead of requiring a developer to edit a dataset manually, the system supports a user-facing path from detection to naming and local persistence. The adaptive object-learning workflow has a similar product value because it allows the robot to acquire household-specific vocabulary over time.

7.5 Limitations and threats to validity

The first limitation is sample size. Five known users and approximately twenty unfamiliar image/video samples are not sufficient for general biometric conclusions.

The second limitation is uncontrolled test conditions. Lighting, camera distance, phone-display quality, and user pose were not standardised, which means outcomes cannot be compared statistically.

The third limitation is recognition consistency. During practical use, an enrolled person could occasionally appear as UNKNOWN. Likely contributing factors include lighting variation, pose, blur, face size, and threshold behaviour. The current recheck and persistence logic reduces repeated false alerts but does not eliminate false rejection.

The fourth limitation is incomplete fall validation. The fall module is implemented, but a credible full-body trial requires a camera view that captures the necessary landmarks. The available development arrangement did not meet this condition. The fifth limitation is deployment maturity: the application uses development/local-network endpoints and some dashboard values remain demonstrative. The sixth limitation is model artefact completeness: placeholder trained-model files in the inspected repository cannot be treated as a reproducible trained classifier binary.

Finally, expression classification should not be interpreted as direct emotional-state measurement. Cultural, individual, contextual, and situational differences can make the same facial movement ambiguous. BUDDY therefore uses the result as a soft interaction cue and should combine it with temporal/contextual evidence in future work.

7.6 Implications

The findings suggest that useful companion-robot visual intelligence can be built from modular, mostly off-the-shelf perception components when they are combined with careful state management and user-facing controls. For student and low-cost robotics projects, this approach may be more practical than training every model from scratch.

The results also highlight that deployment quality depends as much on integration choices as on model selection. Repeated observations, explicit UNKNOWN states, user-confirmed enrollment, graceful fallbacks, and asynchronous processing are

critical when machine-learning outputs are used in a live robot. These principles can generalise to other domestic service robots and interactive embedded systems.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

BUDDY's visual-intelligence layer contributes to a value proposition centred on affordable, local, personalised companionship. Potential target users include households interested in interactive robotics, educational institutions, researchers, technology demonstrators, and eventually users who could benefit from non-clinical monitoring or companionship features. The combination of familiar-person recognition, unknown-person awareness, camera monitoring, and adaptive object memory differentiates the concept from a simple remote-controlled robot.

The current prototype is not ready for safety-critical care deployment. Any commercial positioning should therefore begin with education, research, entertainment, and general household interaction rather than medical monitoring or certified security services.

8.2 Market and competitive position

The project competes conceptually with social robots, pet-like robots, indoor cameras, and smart-home assistants. Its potential advantage is integration: one embodied platform combines movement, offline voice interaction, visual recognition, environmental sensing, and application control. A low-cost architecture based on broadly available components could reduce the entry cost compared with specialised commercial social robots. The main competitive weakness is maturity. Commercial products typically offer polished hardware reliability, secure cloud/app infrastructure, and extensive testing. BUDDY currently demonstrates research feasibility and must improve robustness, packaging, privacy controls, power management, and long-duration operation before commercial comparison would be meaningful.

8.3 Business or sustainability model

A sustainable development model could use a modular base robot with optional software capability packs rather than a high recurring cloud cost. Core recognition and interaction functions can remain local, while optional remote services could be

added only when needed. This architecture supports privacy and reduces dependence on constant connectivity. For an academic/research pathway, the stronger short-term model is a reusable educational platform. Universities and schools could use BUDDY to teach robotics, embedded AI, HRI, computer vision, and ROS 2 integration. Continued open or documented interfaces would allow new student teams to replace individual modules without redesigning the entire robot.

8.4 Scalability, intellectual property and compliance

Scaling the visual subsystem requires configurable network discovery, secure authentication, encrypted storage/transport, user-account permissions, and a structured biometric-data lifecycle. Production deployment must also review the licenses of all third-party libraries and pretrained models, particularly where wrapped frameworks include models with different license terms.

Intellectual-property value in this project lies mainly in the integrated robot architecture, interaction workflows, application design, and project-specific adaptation rather than ownership of foundational models such as MobileNetV2, DeepFace components, or MediaPipe. Commercialisation would therefore require a careful dependency and licence audit.

8.5 Contribution to the Sustainable Development Goals

The work most directly supports SDG 9 (Industry, Innovation and Infrastructure) through experimentation with affordable embedded AI and robotics. It can also contribute indirectly to SDG 3 (Good Health and Well-being) through future non-clinical companionship and safety-support research, provided such functions are validated responsibly. Educational deployment of the platform can support SDG 4 (Quality Education) by providing a hands-on environment for robotics and AI learning.

Sustainability also requires avoiding unnecessary hardware and cloud dependence. Reusing commodity computing platforms, maintaining modular replaceable components, and performing suitable processing locally can extend hardware usefulness and reduce continual network/service requirements.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the individual visual-intelligence contribution to BUDDY, an intelligent multimodal robot-dog. The component evolved from face recognition and facial-expression processing into a broader visual-awareness subsystem after supervisory feedback identified the need for greater research depth. The final scope includes familiar-person recognition, unknown-person handling and enrollment, facial-expression analysis, active camera-head face tracking, adaptive object learning, pose-based fall-detection logic, emotion logging, camera media services, and a Flutter companion application.

Functional evidence demonstrates that the subsystem can recognise enrolled users, reject unfamiliar faces, create application-visible unknown-person events, support explicit household enrollment, analyse live facial expressions, learn and identify a small set of user-taught objects, stream/record camera media, and drive face-following behaviour. The work also identified genuine limitations, including occasional false rejection of known users and incomplete physical fall validation caused by an unsuitable development-camera view.

The key conclusion is that companion-robot visual intelligence should be treated as a layered awareness problem rather than a single classification problem. Practical value emerges when perception is combined with memory, uncertainty handling, user review, adaptive learning, and integration with the rest of the robot.

9.2 Achievement of objectives

Objective	Achievement
Familiar-person recognition and enrollment	Achieved at subsystem level; occasional false rejection remains
Seven-category facial-expression analysis	Achieved in live operation
Pan/tilt face following	Achieved with physical integration support
Unknown-person alert and user-reviewed enrollment	Achieved at subsystem/app level

Objective	Achievement
Adaptive object learning and identification	Achieved for mug, box and phone
Pose-based fall-detection logic	Implemented; final full-body physical validation pending
Camera streaming/recording and application integration	Achieved at subsystem level
Transparent functional evaluation without unsupported accuracy claims	Achieved

Table 9.1 Achievement of individual objectives

9.3 Individual research and product contributions

Sobiya's individual research contribution is the expanded visual-awareness architecture and its practical implementation. The component demonstrates how a low-cost companion robot can combine multiple vision functions while preserving explicit UNKNOWN states and human-reviewed enrollment. Her product contribution includes the primary functional implementation of the Flutter companion app and the backend services required to expose the visual subsystem to a user.

Team contributions are acknowledged where relevant: Yohan provided UI design input, while Ashandth assisted with final physical integration/troubleshooting of the camera-head mechanism. The visual perception, adaptive-learning, and application-backend work remain Sobiya's individual component.

9.4 Future directions

Immediate future work is to complete full-body fall-detection validation using a camera placement that reliably captures the necessary pose landmarks. The fall module should then be evaluated with repeated fall and non-fall scenarios so that false-positive and false-negative behaviour can be quantified rather than described only functionally.

Face recognition should be strengthened through multi-view enrollment, more diverse lighting conditions, temporal identity voting, stronger modern embeddings, and systematic threshold calibration. Unknown-person alert logic can be extended

with event grouping so that one visitor does not generate repeated records while remaining in view. Anti-spoofing should be considered because phone-displayed faces were used during development and a production security feature should distinguish a live person from a presented image where feasible.

Expression-aware interaction should incorporate temporal persistence, recognised-user context, and interaction history instead of reacting to a single frame. Object learning can be extended beyond a central crop by combining lightweight detection with user-taught labels, allowing multiple objects in one scene. Application deployment should replace hard-coded network addresses with configuration or service discovery and add authentication, encrypted transport, and user-controlled data deletion.

Finally, all visual features should be re-evaluated after final integration on the completed Raspberry Pi robot because CPU load, camera placement, movement vibration, lighting, and simultaneous voice/navigation processing can change behaviour compared with laptop-only development.

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GLOSSARY

Adaptive object learning - A process in which the user introduces a new object name and the system stores a visual representation for later similarity-based identification.

Embedding - A numerical feature representation used to compare visual samples in a compact feature space.

False rejection - A case where a genuinely enrolled/known user is classified as unknown.

Facial-expression cue - A model-generated label describing visible facial appearance; it is not treated as a definitive internal emotional state.

Unknown-person event - A persistent visual event created when a detected face cannot be matched to a trusted enrolled identity after configured checks.

APPENDICES

Appendix A Functional validation evidence

This appendix summarises the evidence retained for the individual component. Screenshots supplied during thesis preparation show: (a) a dashboard with recognised-person mood and unregistered-person status; (b) a critical unregistered-person notification with a captured face and registration action; (c) an enrollment dialog containing the unknown face; (d) live camera monitoring and recording controls; (e) a recorded media item available for retrieval; (f) live UNKNOWN identity with facial-expression output; and (g) learned-object identification of a previously taught box.

Additional final-presentation evidence should include a full-body fall-detection demonstration, the final Raspberry Pi camera deployment, and integrated operation while the voice/navigation subsystems are active. These additions can strengthen the final viva evidence without changing the distinction made in this thesis between already demonstrated functions and pending final integration validation.

Appendix B Academic integrity and AI use disclosure

AI-assisted tools were used during the project and thesis preparation. GitHub Copilot was used by Sobiya Anton Suresh as a programming-assistance tool during parts of the Flutter/mobile application and software development process. ChatGPT was used during thesis preparation to assist with structure, language refinement, synthesis of supplied project evidence, and preparation of draft academic prose.

AI-generated suggestions were not treated as evidence by themselves. Project-specific technical claims were checked against the BUDDY research paper, PP2 presentation, source-code repositories, supplied screenshots, and direct functional-test information provided by the project team. Unsupported statistical accuracy values were intentionally excluded. Final responsibility for reviewing, correcting, and submitting the thesis remains with the student.

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